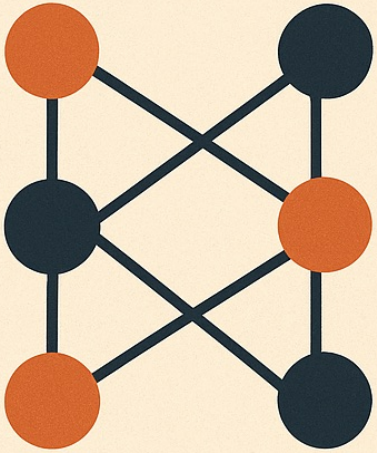
INTRO TO MACHINE LEARNING

AND NEURAL NETWORKS





Lecture 1: Intro to classical ML

#Agenda

1) Recap of PCA
2) PCA [Continued]
3) What is ML?
4) Types of ML Algorithm

Recap

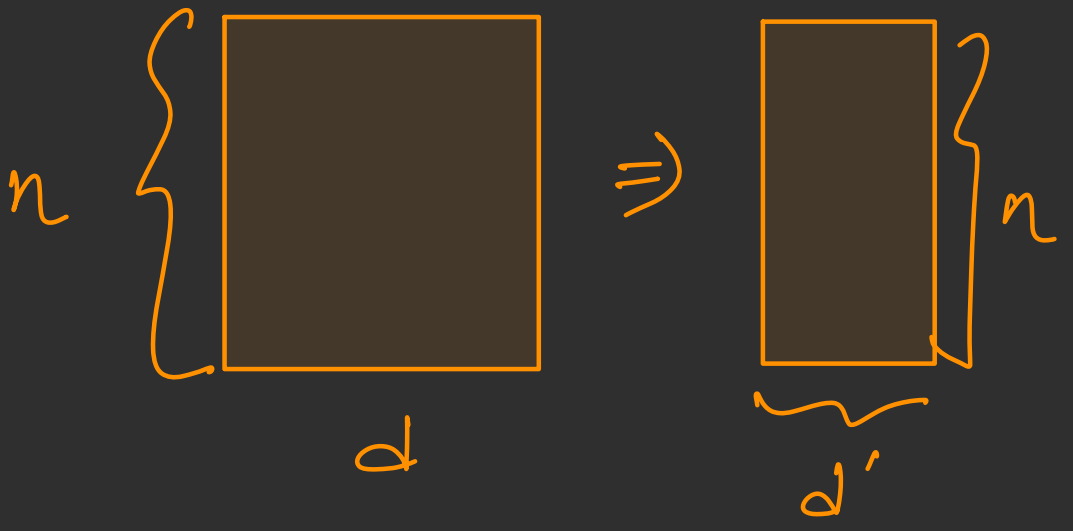
{ Reduce Dimension }

↳ Col / features

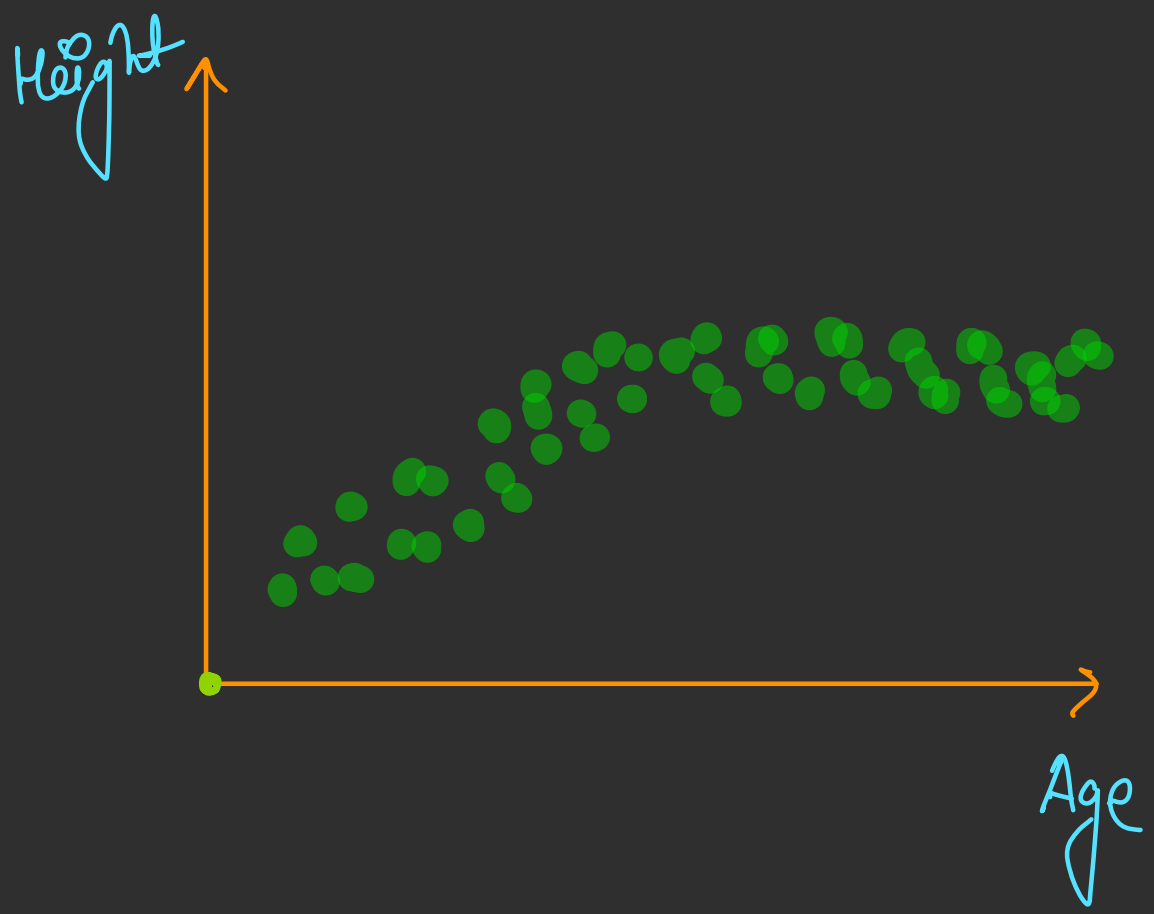
1) Principal Component Analysis

$$D\left\{x_{i,j}, x \in R^d\right\}_{i=1}^{n} \longrightarrow \boxed{\text{PCA}} \longrightarrow D'\left\{x_{i,j}, x \in R^{d'}\right\}_{i=1}^{n}$$

1. $d == d'$
2. $d >> d'$ ✓
3. $d << d'$



Example: Height Vs Age



-1 to 1 ⇒

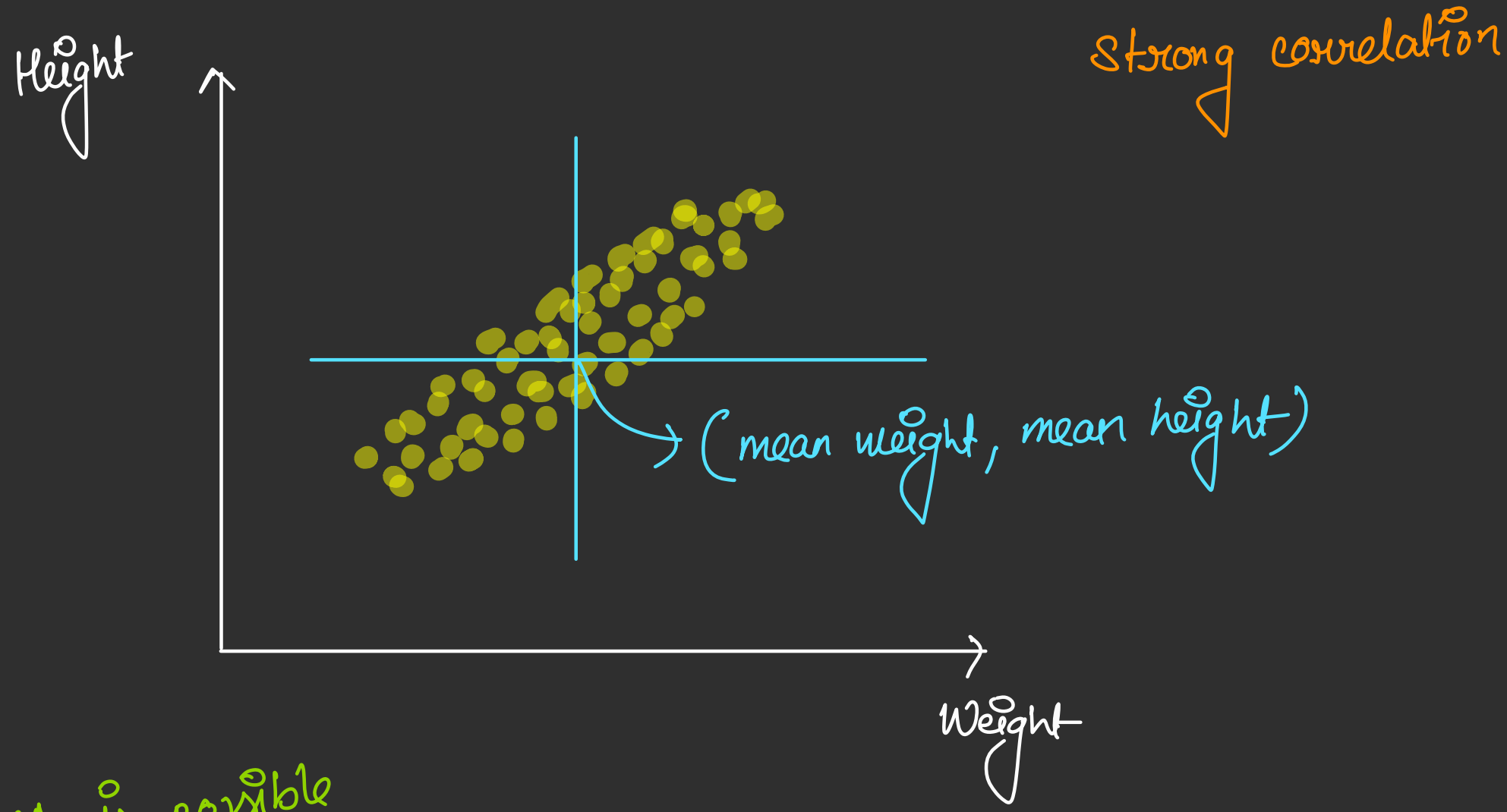
Pearson Corr

Weak correlation

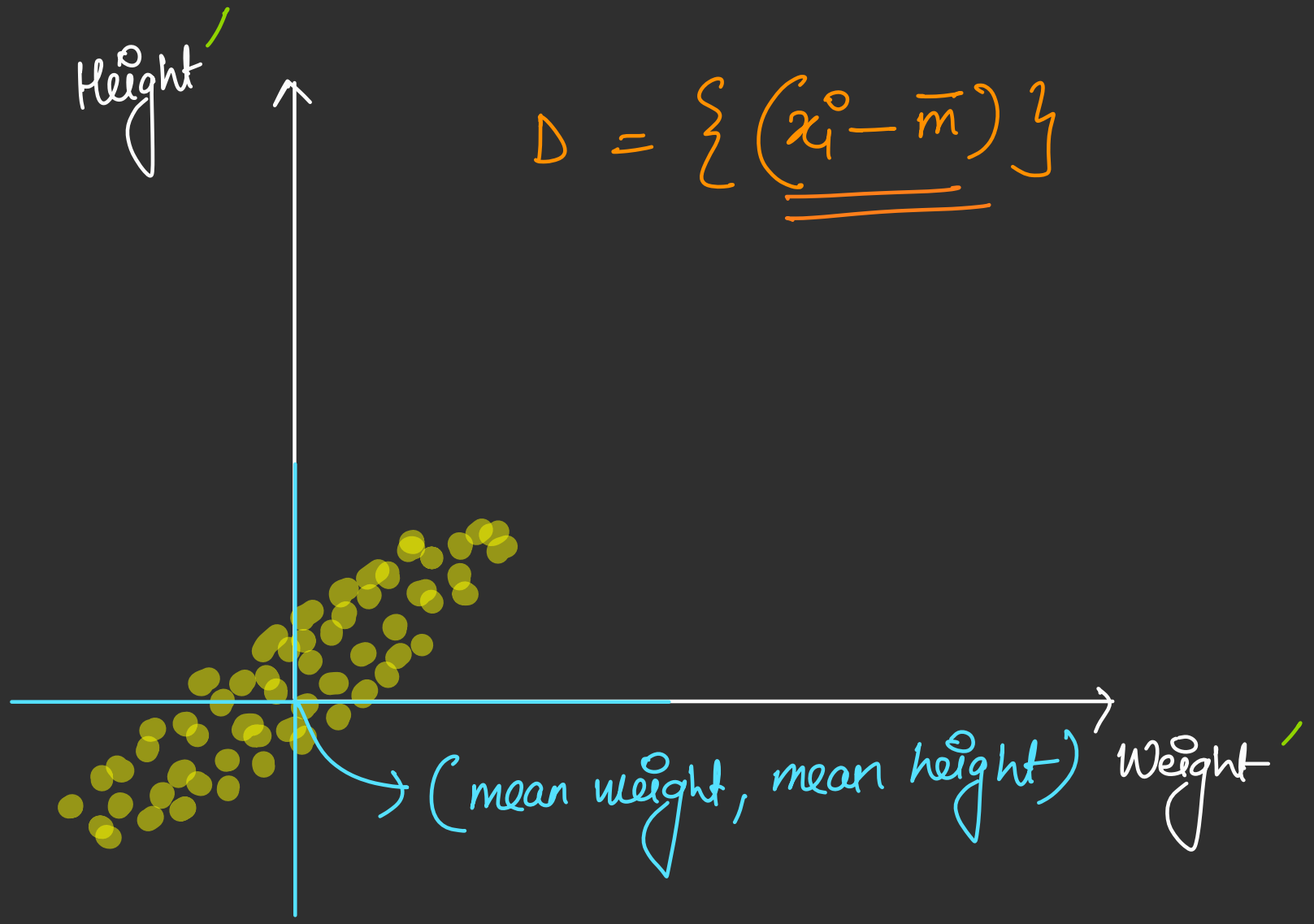
↓

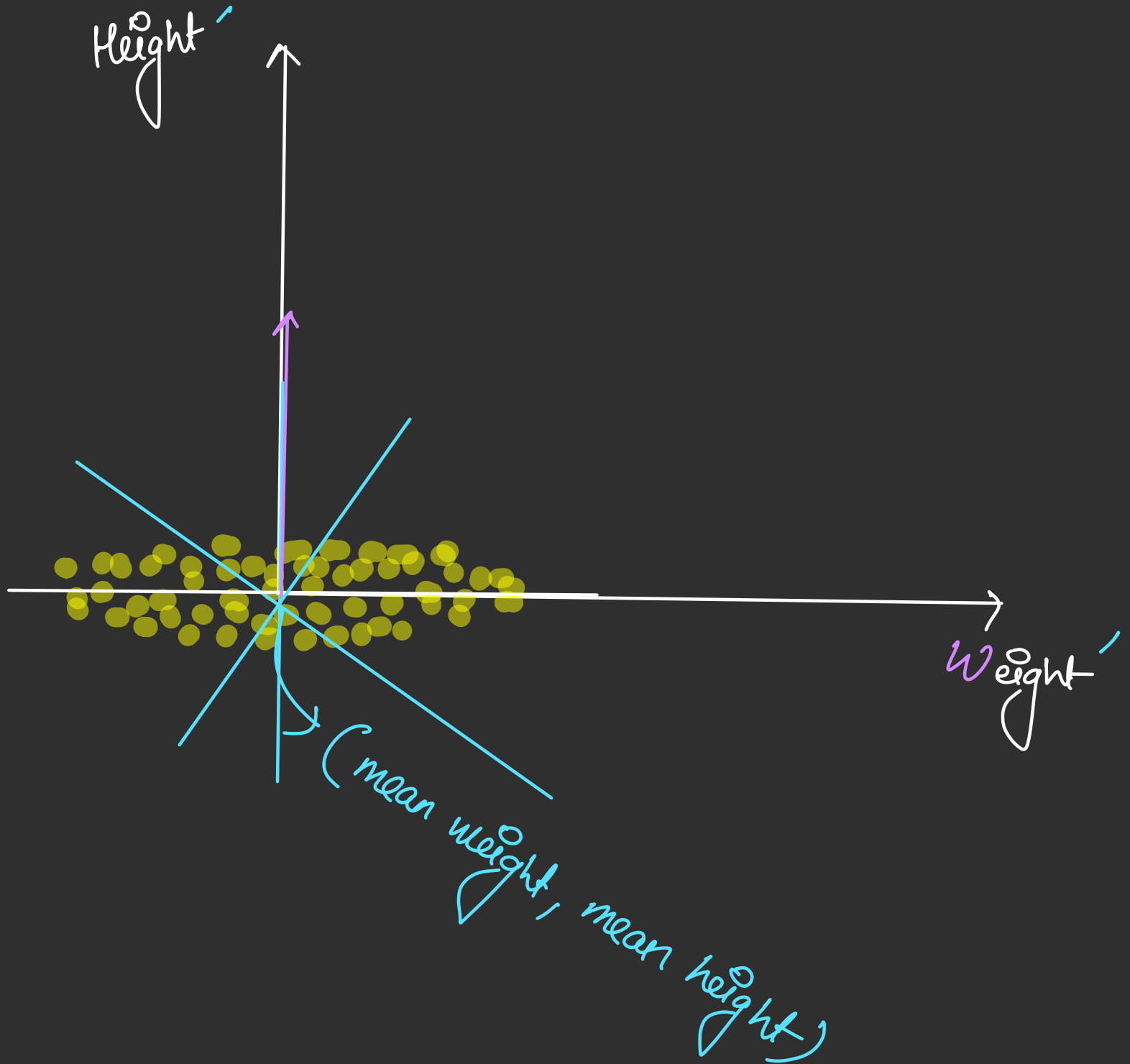
DR

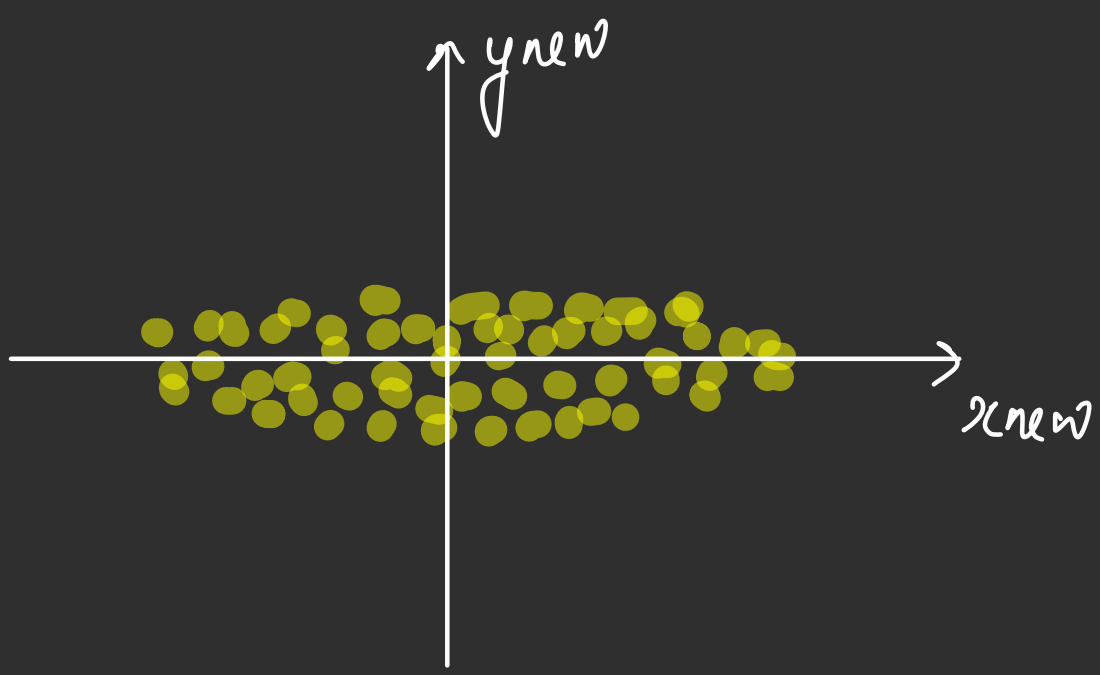
« Not possible in this case »

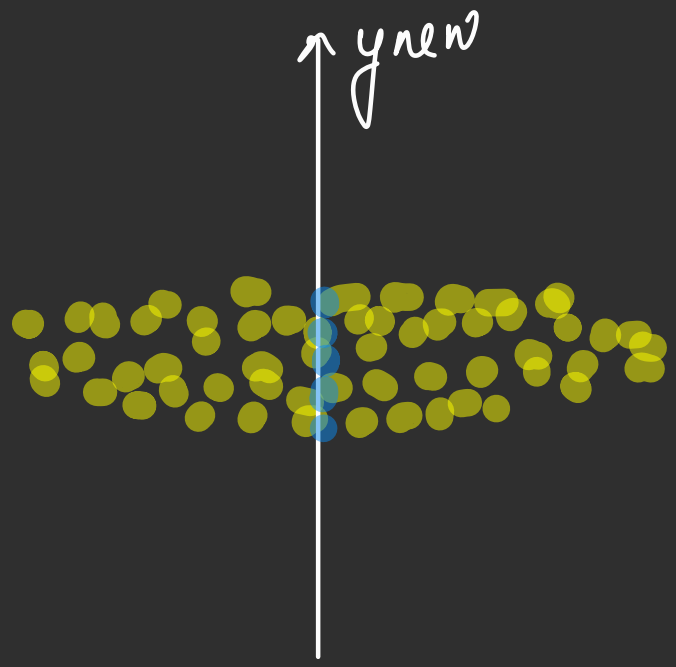


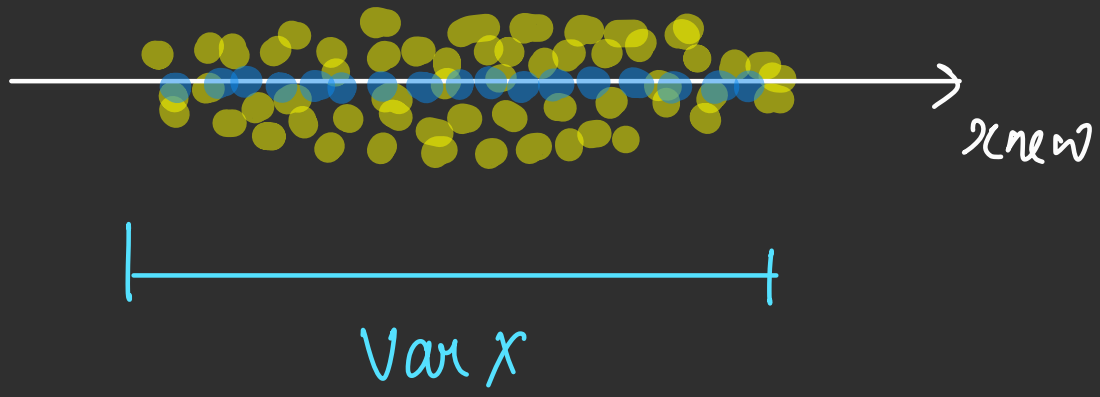
PCA is possible

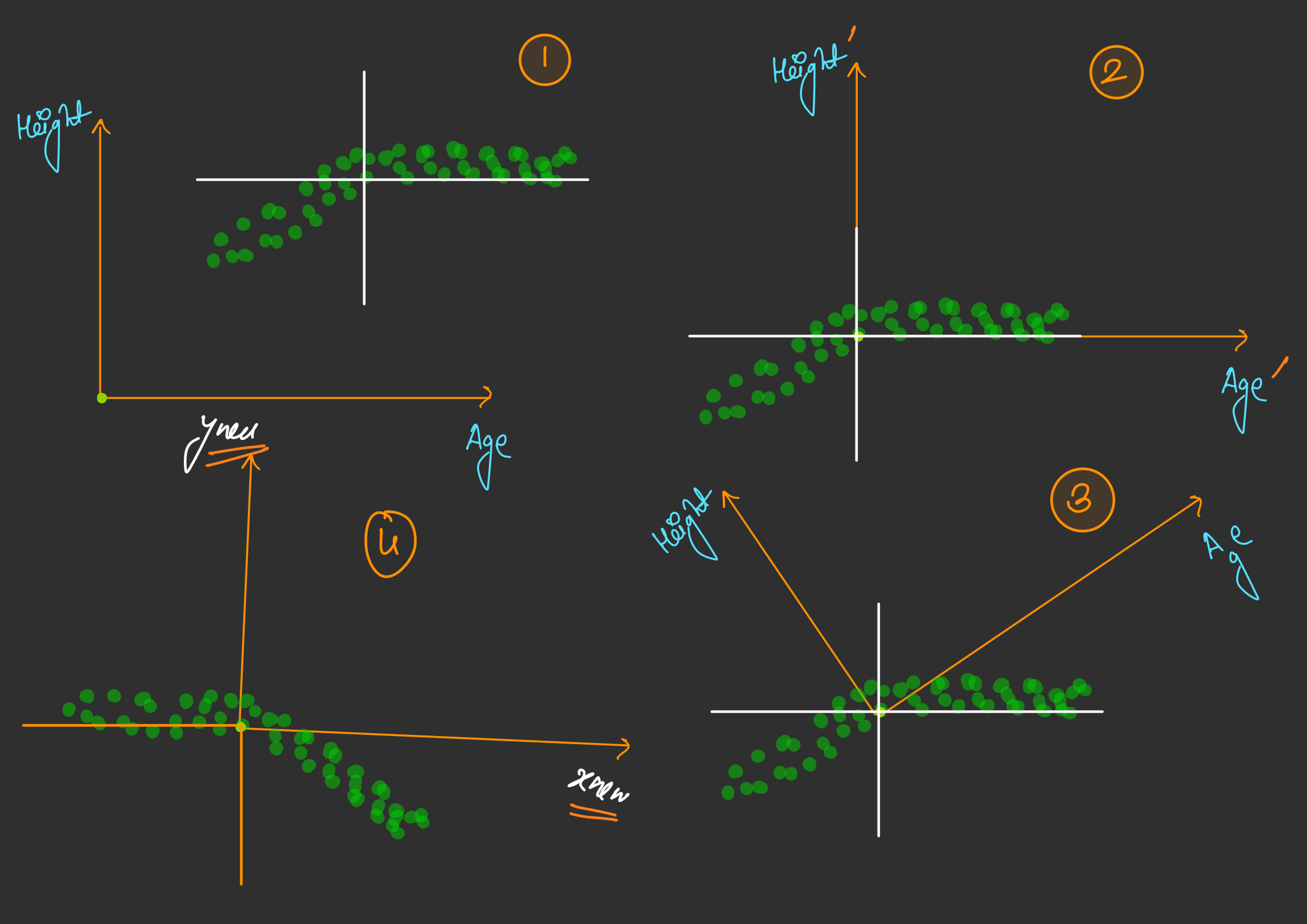

Height
$D = \{(x_i - \bar{m})\}$
(mean weight, mean height)
Weight


Height'
Weight'
(mean weight, mean height)


ynew
xnew


ynew


xnew
Var x

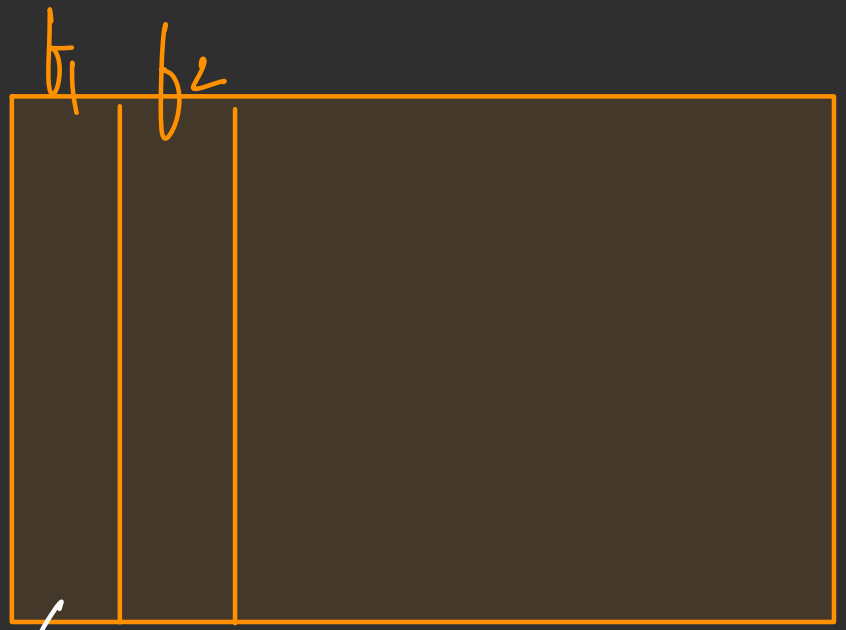

1
Height
Age
2
Height'
Age'
3
Height
Age
4
ynew
xnew

1) Find the new axes → 2 axis (x_{new}, y_{new})

2) Where var is max → x_{new}

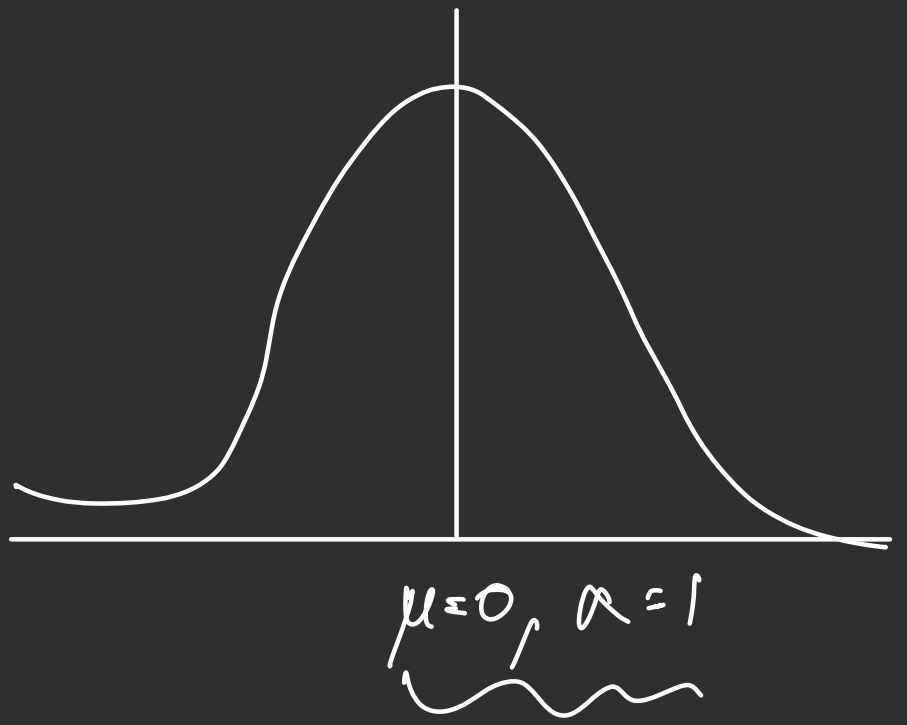
Implementation PCA

① Column Standardization



→ $\bar{x}, \sigma$

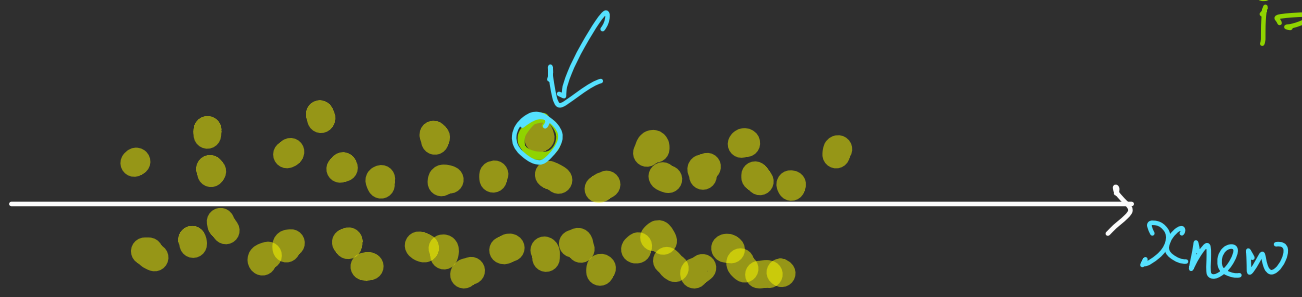
$\left(\frac{x_i - \bar{x}}{\sigma}\right)$ → Z score

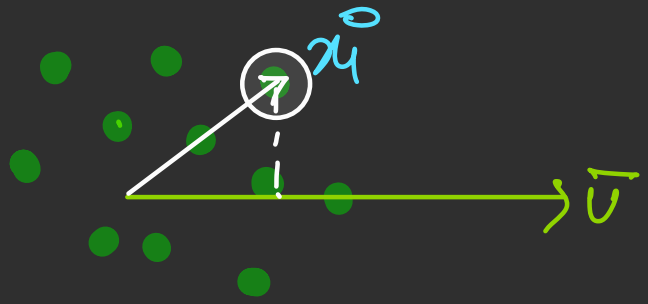


"Standard Normal"

(2) Find a vector $\bar{u}$ such that the variance of $\bar{x}$ along $\bar{u}$ is maximum

$$Var = \frac{\sum_{i=1}^{n} (x_i - \mu)^2}{n}$$





Projection of x_i on $\bar{u}$ $\Rightarrow \underbrace{\frac{x_i \cdot \bar{u}}{\|u\|}} \Rightarrow x_i \hat{u}$

$$Var(x_i) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{x_i \bar{u}}{\|u\|} - \mu \right)^2$$

$\mu \rightarrow 0$

$$\text{Var}(x_i^o) = \frac{1}{n} \sum_{i=1}^{n} \left(\frac{\bar{x}_i^{oT} \bar{u}}{\|u\|} \right)^2 \longrightarrow \text{fun}^n \text{ of } \bar{u}$$

↳ max

(Applying Lagrange's Multiplier)

$$\bar{u}^* = \underset{\bar{u}}{\text{argmax}} \frac{1}{n} \sum_{i=1}^{n} \left(\bar{x}_i^{oT} \cdot \bar{u} \right)^2 + \lambda \left(\|u\| - 1 \right)$$

Lets Continue

$$\bar{u}^* = \underset{\bar{u}}{\operatorname{argmax}} \frac{1}{n} \sum_{i=1}^{n} \left(\bar{x}_i^T \cdot \bar{u} \right)^2 + \lambda \left(\|u\| - 1 \right)$$

$$n \left\{ \begin{bmatrix} \text{---} & \bar{x}_1 & \text{---} \\ \text{---} & \bar{x}_2 & \text{---} \\ \text{---} & \bar{x}_3 & \text{---} \\ \text{---} & \bar{x}_n & \text{---} \end{bmatrix}_{n \times d} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_2 \\ \bar{u}_3 \\ u_n \end{bmatrix}_{d \times 1} \Rightarrow \begin{bmatrix} x_1^T u_1 \\ x_2^T u_2 \\ \vdots \\ x_n^T u_n \end{bmatrix} \right\} = \|u\|^2$$

d

This matrix is called a data matrix or X

$$\bar{u}^* = \underset{\bar{u}}{\operatorname{argmax}} \frac{1}{n}\left(\bar{u}^T x^T x \bar{u}\right) + \lambda\left(\|\bar{u}\|^2 - 1\right)$$

$$(A^T B)^T \Rightarrow B^T A$$

$$(A^T B)^2 \Rightarrow B^T A^T A B$$

↳ property of matrix

$$(x^T u)^2$$

$$\bar{u}^* = \underbrace{\underset{\bar{u}}{\operatorname{argmax}} \frac{1}{n}\left(\bar{u}^T x^T x \bar{u}\right) + \lambda\left(\|\bar{u}\|^2 - 1\right)}_{f(\bar{u})}$$

$$\vec{v} = x\bar{u}$$

$$\bar{v}^T\bar{v} = (x\bar{u})^T(x\bar{u})$$

$$\Rightarrow \bar{u}^T x^T x \bar{u}$$

$$\bar{u}^T \cdot \underbrace{x^T \cdot x} \cdot \bar{u}$$

$$x = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}_{2\times 2} \Rightarrow x^T \cdot x \Rightarrow$$

$$x^T = \begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}_{2\times 2}$$

matrix (2×2)

Fact

$$f(\bar{u}) = \bar{u}^T \cdot S \cdot \bar{u}$$

Here S can be of any size (matrix)

$$\nabla_{\bar{u}} f(\bar{u}) = (S + S^T) \cdot \bar{u}$$

$$f(u) = \frac{1}{n} u^T \cdot x^T \cdot x \bar{u} + \lambda (u^T u - 1)$$

$\|u\|^2 \hookrightarrow u^T \cdot u$

$$\nabla_{\bar{u}} f(\bar{u}) = \frac{1}{n} \left((x^T x) + (x^T x)^T \right) \bar{u} + \lambda (2\bar{u})$$

$$= \frac{1}{n} \left(x^T x + x^T x \right) \bar{u} + \lambda (2\bar{u})$$

$$\Rightarrow \frac{2}{n} x^T x \bar{u} + \lambda (2\bar{u})$$

Now we have to find the minima,

$$\nabla_{\bar{u}} f(\bar{u}) = 0$$

$$\frac{2}{n} X^T X \bar{u} + \lambda(2\bar{u}) = 0$$

$$\Rightarrow 2\left\{\frac{1}{n} X^T . X \bar{u} + \lambda \bar{u}\right\} = 0$$

$$\Rightarrow \frac{1}{n} X^T . X . \bar{u} = -\lambda \bar{u}$$

$$\Rightarrow X^T X \bar{u} = -n\lambda \bar{u}$$

$$\Rightarrow \boxed{X^T X \bar{u} = \lambda' \bar{u}}$$

The vector $\bar{u}$ which maximizes the fun^n $f(\bar{u})$ satisfies Eq (1)

$$X^T \cdot X \; \bar{u} = \lambda' \bar{u}$$

$\bar{u}$ → Eigen vector of matrix $X^T X$

Eigen → german word for Identity

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

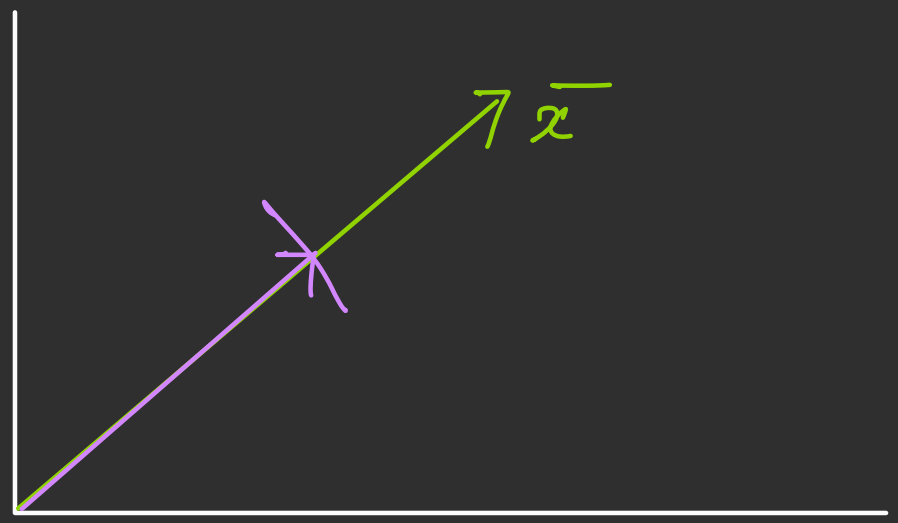
For any square matrix S

square → No of rows = No of cols

if we can find $\bar{x}$ such that

$$S\bar{x} = \lambda \bar{x}$$

$\bar{x}$ is said to be eigen vector.



Eigen vector

$S_{d \times d} \cdot \bar{x}_{d \times 1}$

$\lambda \bar{c}$

Step2: The vector $\vec{u}$ which captures more var is the eigen vector of matrix

Numpy

Each matrix has many eigen vectors

Each eigen vector has its own eigen values

We pick the eigen vector correspoinding to the maximum eigen value for our $\vec{u}$.

$$\begin{bmatrix} 2 & 4 \\ 3 & 5 \end{bmatrix}$$

$$\vec{\mu} = [20, 21]$$

max

$\Rightarrow \quad d \rightarrow d' \qquad\qquad d' < d$

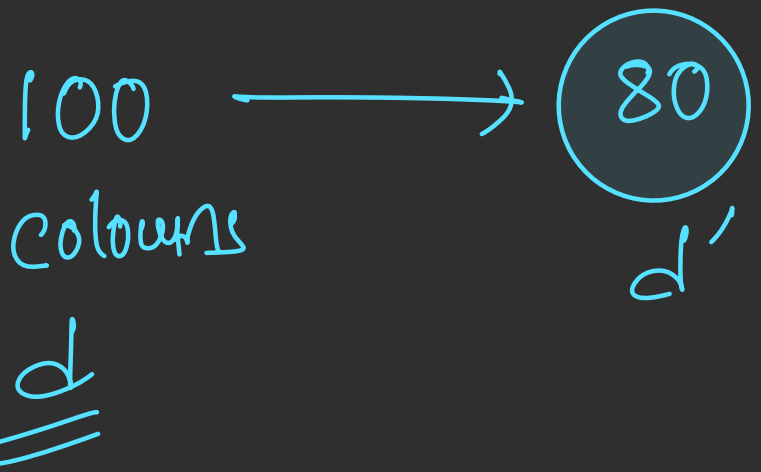
We simply pick the first d' eigen vectors with largest eigen values.

Step 3: Convert $\bar{x_i} \in R^d$ to $\bar{x_i} \in R^{d'}$

$$\frac{\bar{x_i}^T \cdot \bar{u_1}}{\|u_1\|} + \frac{\bar{x_i}^T \cdot \bar{u_2}}{\|u_2\|} + \frac{\bar{x_i}^T \cdot \bar{u_3}}{\|u_3\|} \cdots \quad d'$$

→ Coordinate along $\bar{u_1}$

→ Coordinate along $\bar{u_2}$



1) Col standardization

2) $X^T \cdot X$

3) Eig vector
   ↳ numpy

4) Max(Eig values)
   ↳ Eig vectors

Intro to ML and NN

Machine Learning

80%

Neural Networks

20%

1) ML & NN → Linear Reg & Logistic Reg

2) Supervised ML

3) Unsupervised ML

4) Neural Network → CV & NLP

1) Python, Python lib
2) Prob & Stat
3) Maths for ML

} Fundamental & pre-requisite

Q) What is Machine learning?

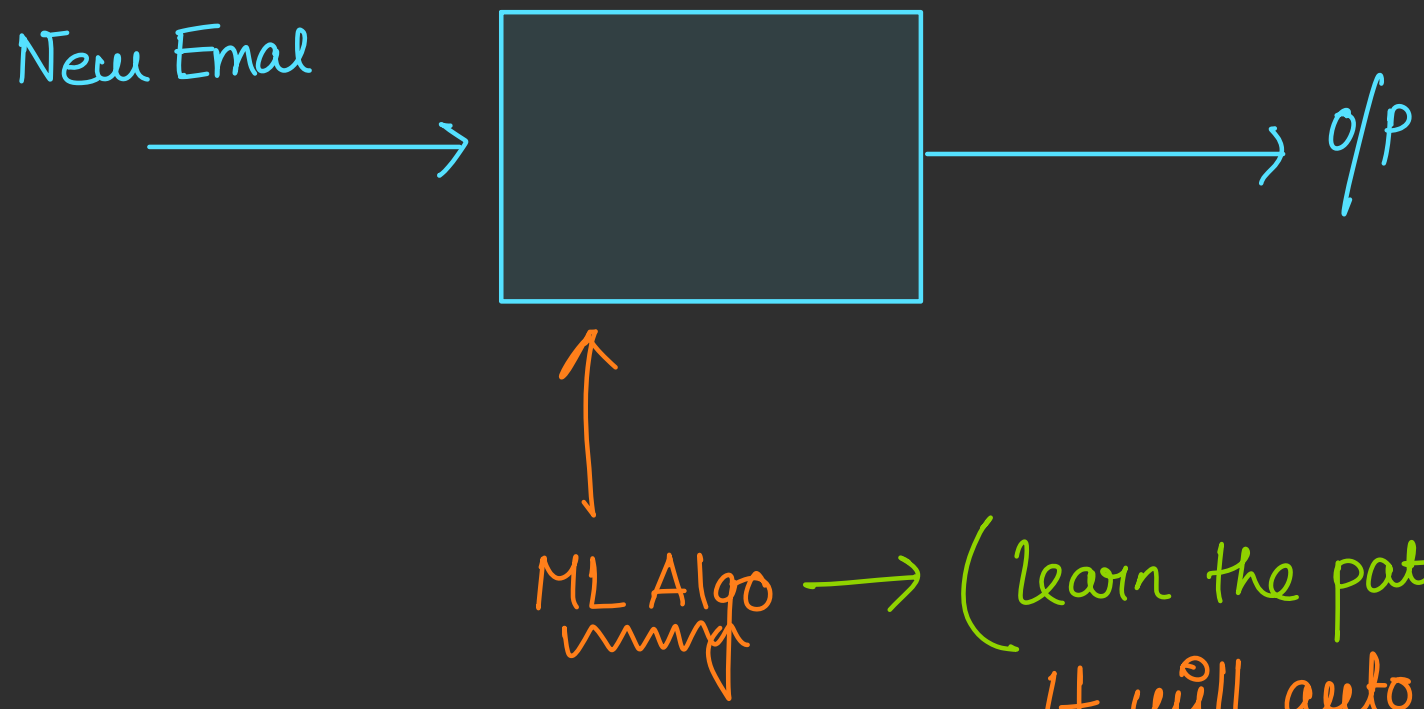
gmail → Ham

gmail → Spam

Rules → if else condition → filter

Rule → hard code




New Email
o/p
ML Algo
(learn the pattern in the data)
It will automatically learn
the rules

Machine Learning

Algo

→ Learning the hidden pattern in my data

Task

Clarify SPAM Vs HAM

100 → New Email

Experience

Historical data

Performance

Quantifie way to measure sucess

60 / 100 => Accuy

60%

Task

Exp

Performance

Predict stock price

Segment my customer

Catorize cat & dog

ML prob Statement

Classification

Dog Vs Cat, Spam Vs Ham

Mango Vs Orange

Regression

predict the sales value for the nxt quarter

"House prices"

Clustering

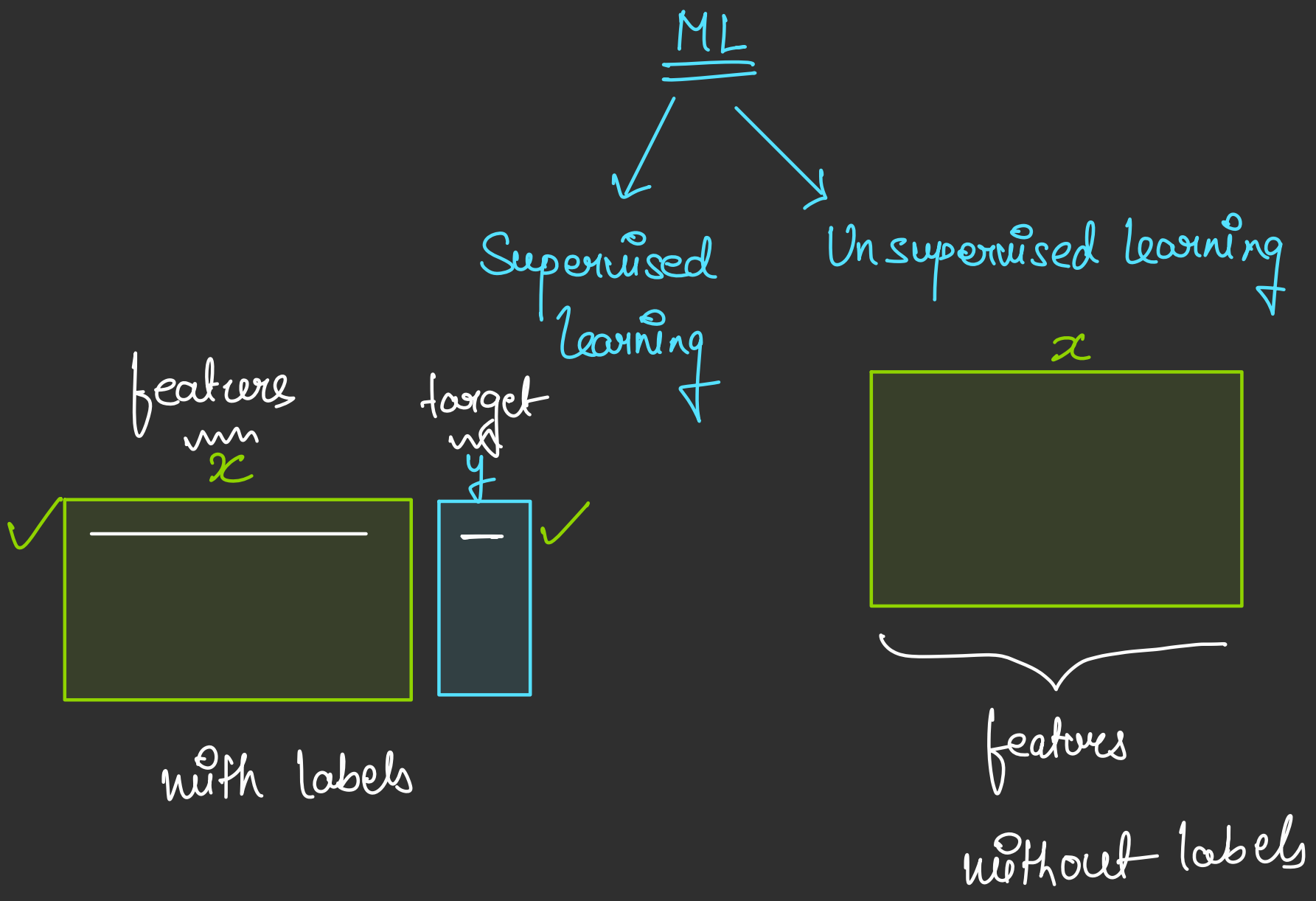
Segment my customer to understand whom to give discount

Recommedation

Youtube, Netfix

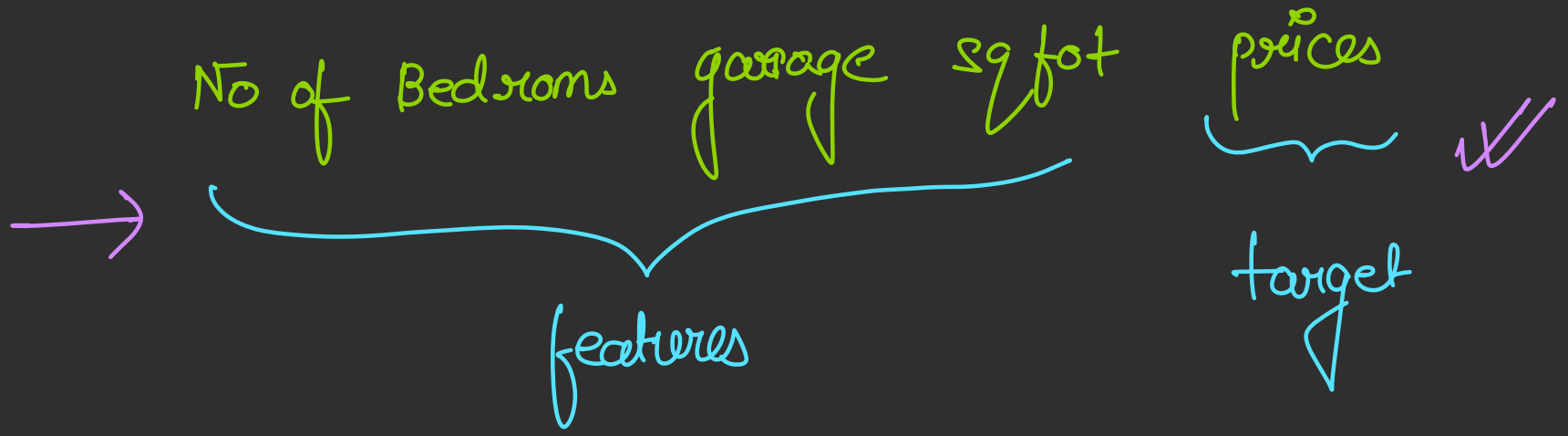
Forecasting

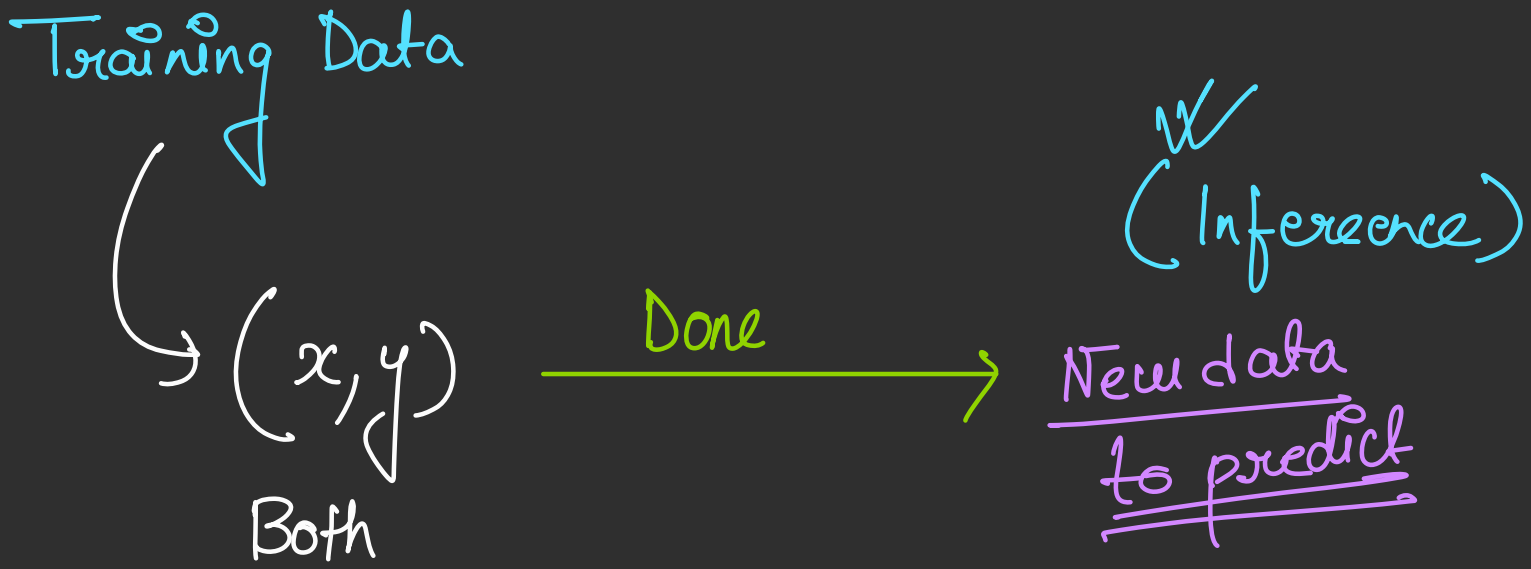
predict of stock prices for the next year


ML
Supervised learning
Unsupervised learning
features
x
target
y
with labels
x
features
without labels

Supervised learning

Predict of House price




Training Data
(x, y)
Both
Done
New data
to predict
(Inference)